Uganda Food Price Prediction

Predictive Analytics for Food Security

Data Engineering & Analytics Project

WFP Uganda Market Data | 2007 - 2026 | 43 Markets | 10,000+ Records

Presented by: Daisy

Project Goal & Business Insight

Objective

Predict market-level food prices across Uganda using historical WFP data

Target Audience

NGOs, Government planners, and Market analysts

Business Value

Enables early action for price spikes, improving food security planning and aid distribution accuracy

Data Integrity & Engineering

Source

WFP (World Food Programme) — 10,000+ rows of verified Uganda market data

Data Cleaning

Strict removal of non-food items (Fuel, Bins, Clothing) to ensure a Food-Only focus

Initial Records: 32171 l Cleaned Records (Food Only): 28903 l Date Range: 15/01/2006 to 15/12/2025 I Rows Removed: 3268

Quality Control

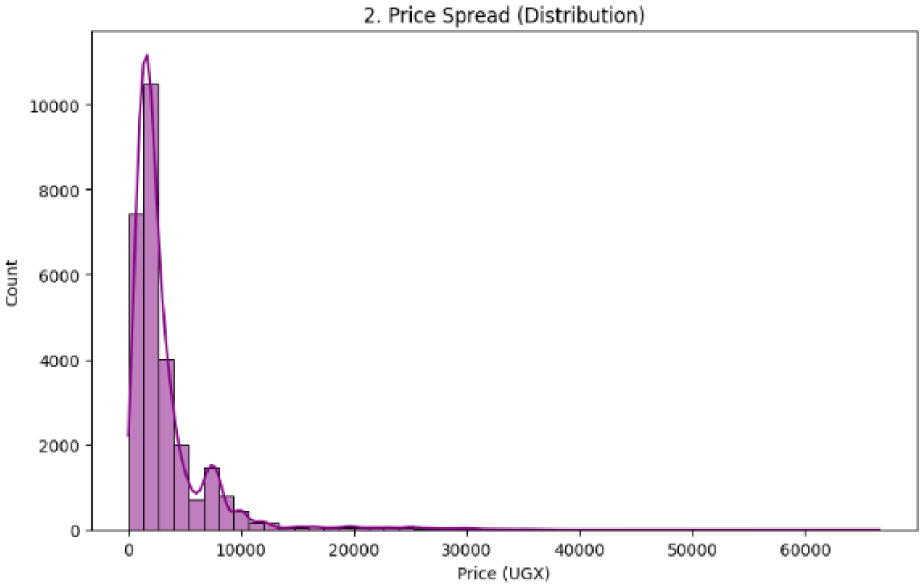
Managed missing values and removed duplicates to maintain model reliability

Automation

Built a pipeline to handle date conversions and geographical mapping automatically

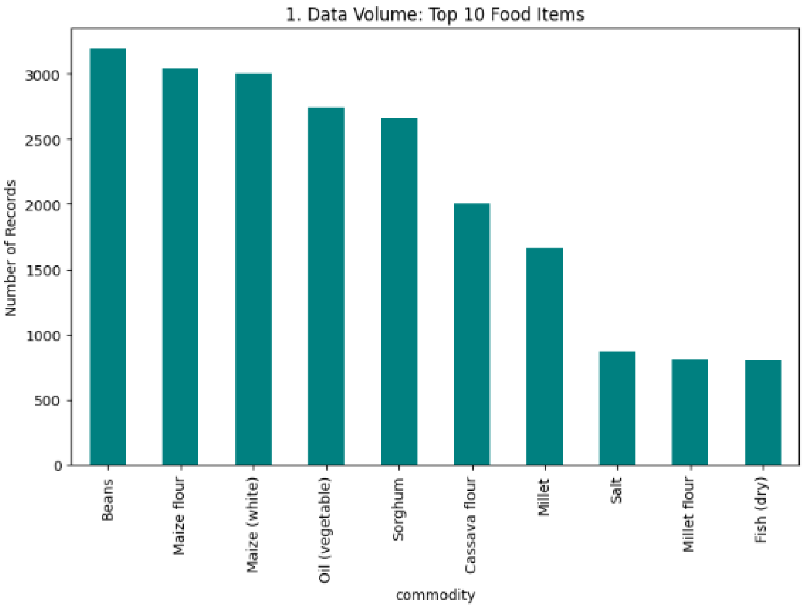
Data Exploration — Univariate Analysis

Price Distribution



Most food prices in Uganda fall below UGX 10,000. The distribution is right-skewed, meaning a small number of expensive commodities pull the average up.

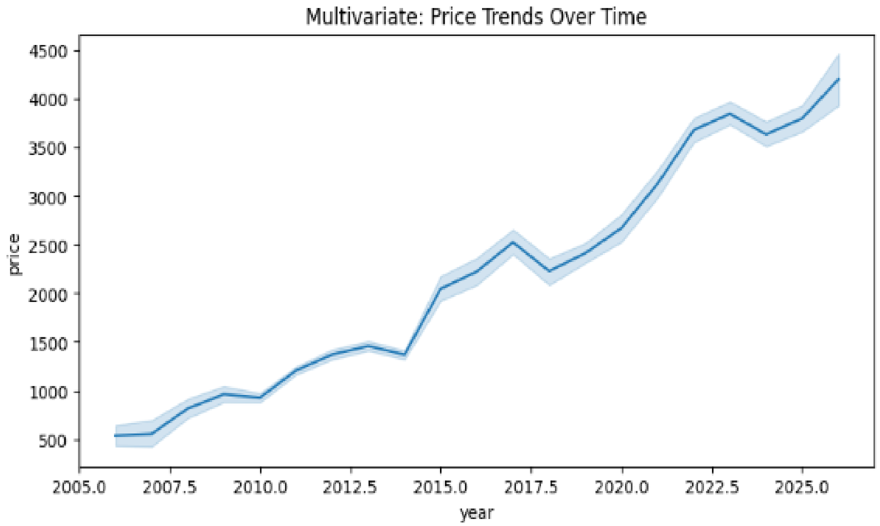
Top 10 Food Items by Frequency



Beans, Maize flour, and Maize (white) are the most frequently traded food items across Uganda's markets, making them the most important commodities to predict.

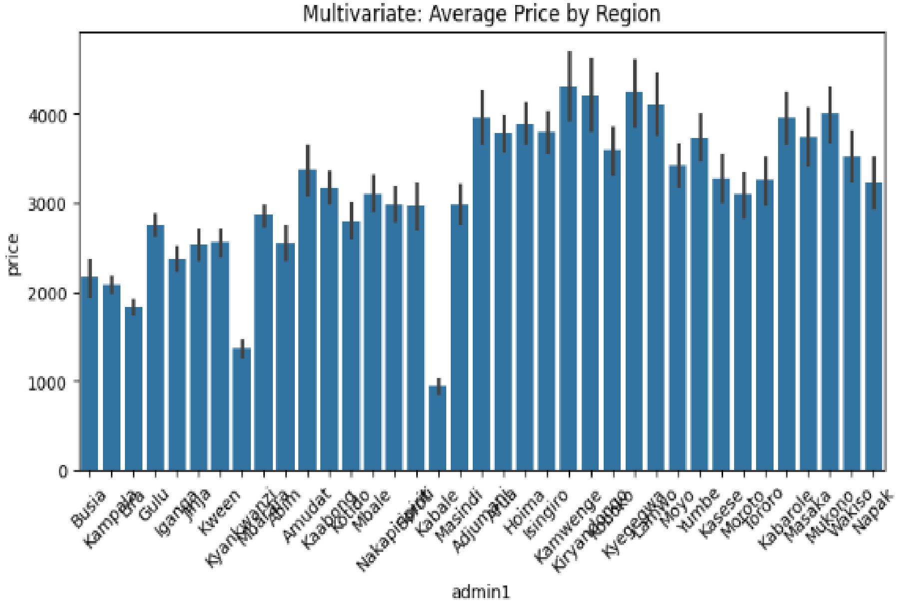
Data Exploration — Multivariate Analysis

Price Trends Over Time



Food prices have risen steadily from around UGX 500 in 2007 to over UGX 4,000 in 2026. There is a sharp rise after 2020, likely linked to COVID-19 disruptions and inflation.

Average Price by Region



Prices vary significantly across Uganda's 43 markets. Some border and northern markets show higher average prices, confirming that location is a strong predictor of food price.

The Machine Learning Approach

Feature Engineering

Created inputs for Year, Month, Region, Market and Commodity

Classical Model

Random Forest Regressor — captures non-linear market behaviours effectively

Neural Network

MLP Regressor — a deep learning approach to identify complex long-term price patterns

Requirement Met

The Classical Random Forest provided a reliable baseline for our tabular data, while the Neural Network allowed us to explore more complex, non-linear relationships in food price volatility.

Model Evaluation Results

Random Forest

Classical ML Model

94.15%

R^2 Accuracy Score

Best model for stable local market predictions

Neural Network (MLP)

Deep Learning Model

79.16%

R^2 Accuracy Score

Requires additional tuning for this dataset

Metric Used: R-Squared (R^2) Score | Random Forest provided the most stable and reliable predictions for Uganda market data

Model Evaluation — Detailed Metrics

Random Forest — 94.15% R2

Neural Network (MLP) — 79.16% R2

R2 Score

RF: 0.9415

MLP: 0.7916

Explains how much of the price variation the model captures. Closer to 1.0 is better.

MAE

RF: Lower is better

MLP: Higher than RF

Mean Absolute Error — the average difference in UGX between predicted and actual price.

RMSE

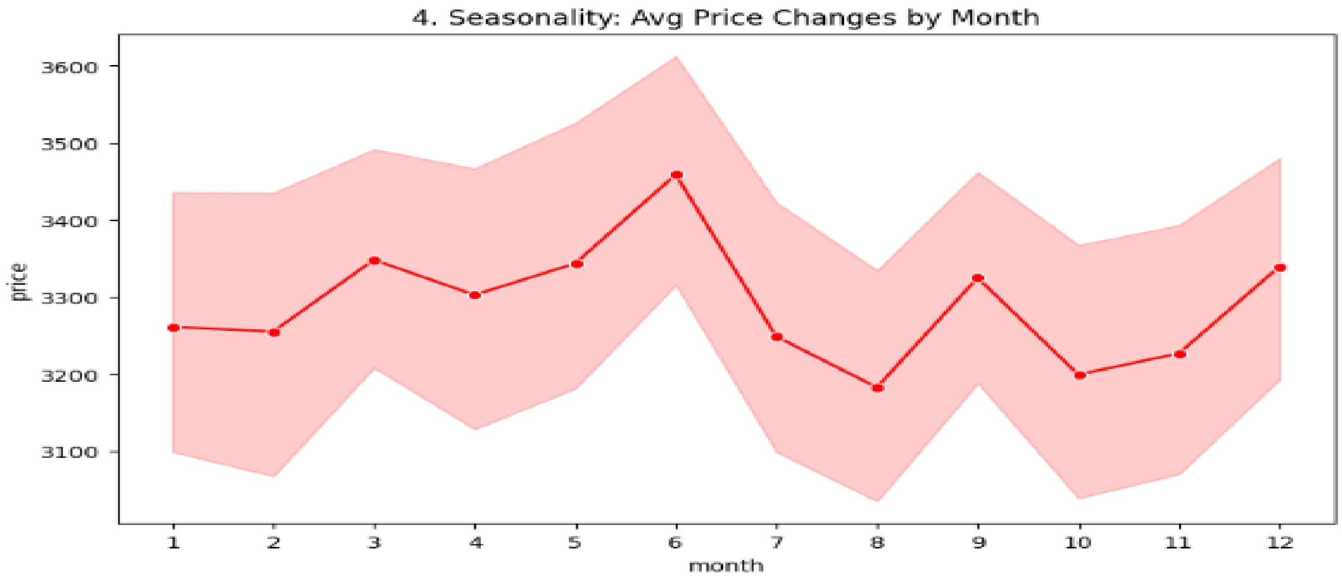
RF: Lower is better

MLP: Higher than RF

Root Mean Squared Error — penalises larger prediction errors more heavily than MAE.

Seasonality Analysis — Average Price by Month

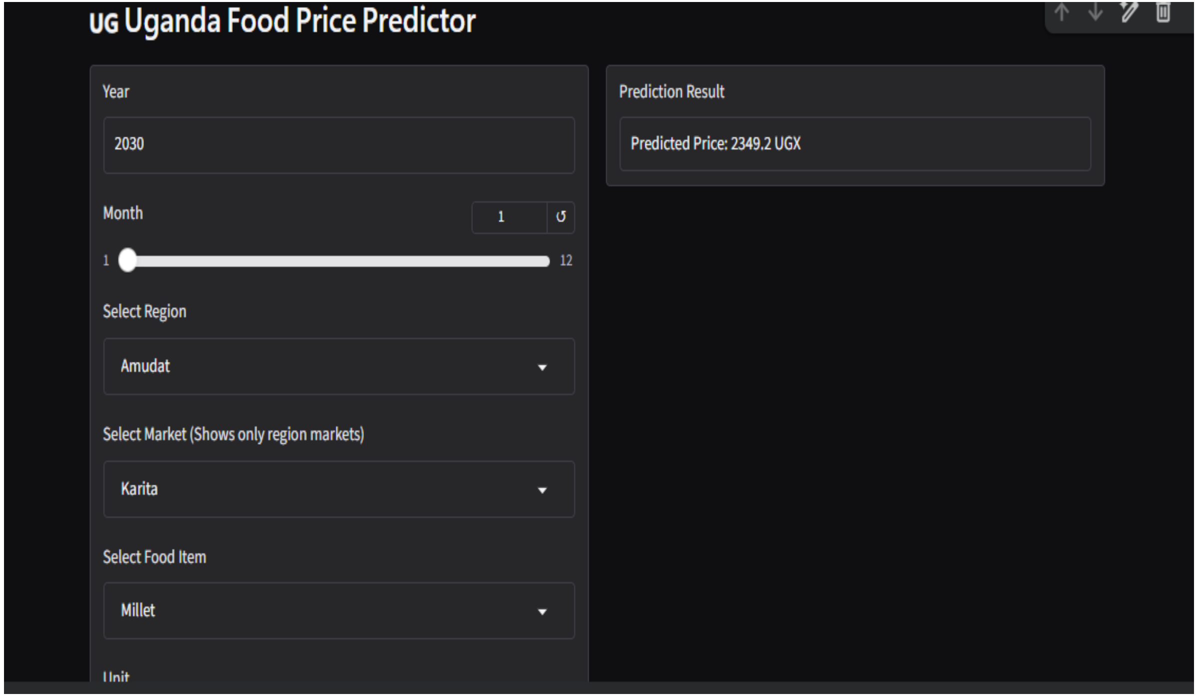
Chart: Average Price Changes by Month



What This Chart Tells Us:

Food prices in Uganda follow a seasonal pattern throughout the year. Prices rise around June and July during the dry season when harvests are lower and food supply drops. They dip between August and October during the second harvest season when more food enters the market. This confirms that Month is a

Live Deployment: Gradio Interface



User Interface

Clean web dashboard built with Gradio

Smart Filtering

Selecting a Region filters only its markets automatically

Accessibility

Designed for non-technical users to get instant price forecasts

Conclusion & Future Work

Summary

Successfully built a robust tool that addresses gaps in Uganda-specific market price predictions

Innovation

Specifically isolated food products to remove economic noise from fuel and labour prices

Next Steps

Integrating climate and rainfall data to improve harvest-season price predictions